LOCAL COMPLETE INTERSECTION HOMOMORPHISMS ARE PERFECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A local complete intersection homomorphism presents the target as a quotient of a polynomial algebra by a Koszul ideal. recalled at proof lines 21857-21860. Sets up the algebraic presentation.
- The ring map is perfect iff the ideal is perfect as a module, by the perfect ring-map and perfect module lemmas. invoked at proof lines 21861-21862. Reduces to perfectness of I.
- Perfectness of a module is local, by the glueing lemma. invoked at proof lines 21863-21864. Allows localizing the Koszul ideal.
- Locally a Koszul ideal is generated by a Koszul-regular sequence, which admits a finite free resolution. invoked at proof lines 21864-21866. Concludes.

Source reference. Original definition 07D0: sources/stacks-more-algebra/pinned-original-source.tex, source lines 8338–8338. Cited at proof lines 21858-21858 to unfold the hypothesis; the reduction and conclusion are inside the counted span.

Source reference. Original lemma 068Y: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21774–21774. Cited at proof lines 21861-21861 to reduce to the ideal; the localization and conclusion are inside the counted span.

Source reference. Original lemma 066Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18917–18917. Cited at proof lines 21861-21862 in the reduction; the conclusion is inside the counted span.

Source reference. Original lemma 066Y: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19077–19077. Cited at proof lines 21863-21863 to localize; the Koszul-regular conclusion is inside the counted span.

Source reference. Original definition 07CV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 8101–8101. Cited at proof lines 21865-21865 to identify the local generator sequence; the finite free resolution conclusion is inside the counted span.

Lemma 0.2. *A local complete intersection homomorphism is perfect.*

Proof. Let $A \to B$ be a local complete intersection homomorphism. By Definition 07D0 this means that $B = A[x_1, \ldots, x_n]/I$ where I is a Koszul ideal in $A[x_1, \ldots, x_n]$. By Lemmas 068Y and 066Q it suffices to show that I is a perfect module over $A[x_1, \ldots, x_n]$. By Lemma 066Y this is a local question. Hence we may assume that

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I is generated by a Koszul-regular sequence (by Definition 07CV). Of course this means that I has a finite free resolution and we win. $\square$